

CSA1715-ARTIFICIAL INTELLIGENCE

LAB PROGRAM- 16

06-08-2026

16. Write the python program to implement Feed Forward Neural Network.

Aim

To implement a Feed Forward Neural Network (FFNN) using Python for classification. The program trains a neural network using given input features and target values, accepts new input from the user, and predicts the corresponding output.

Algorithm

1. Start the program.
2. Define the input training dataset and target output values.
3. Create a Feed Forward Neural Network using MLPClassifier.
4. Specify the number of neurons in the hidden layer.
5. Train the neural network using the training dataset.
6. Provide a test input to the trained network.
7. Perform forward propagation through the network.
8. Obtain and display the predicted output.
9. Create nodes representing the input, hidden and output layers.
10. Connect the nodes using NetworkX.
11. Generate the neural-network architecture using Matplotlib.
12. Stop the program.

Program

```
from sklearn.neural_network import MLPClassifier
import networkx as nx
import matplotlib.pyplot as plt
```

```

X = [[0,0],[0,1],[1,0],[1,1]]
y = [0,0,0,1]

m = MLPClassifier(hidden_layer_sizes=(3,),
                  max_iter=3000,
                  random_state=1)

m.fit(X,y)

x = [1,1]

print("Input:",x)
print("Predicted Output:",m.predict([x])[0])

G = nx.DiGraph()

I = ['X1','X2']
H = ['H1','H2','H3']

for i in I:
    for h in H:
        G.add_edge(i,h)

for h in H:
    G.add_edge(h,'Y')

p = {'X1':(0,1),'X2':(0,-1),

```

```
'H1':(1,1),'H2':(1,0),'H3':(1,-1),  
'Y':(2,0)}
```

```
nx.draw(G,p,with_labels=True,node_size=1800,  
        node_color="lightblue",arrows=True)
```

```
plt.title("Feed Forward Neural Network")  
plt.show()
```

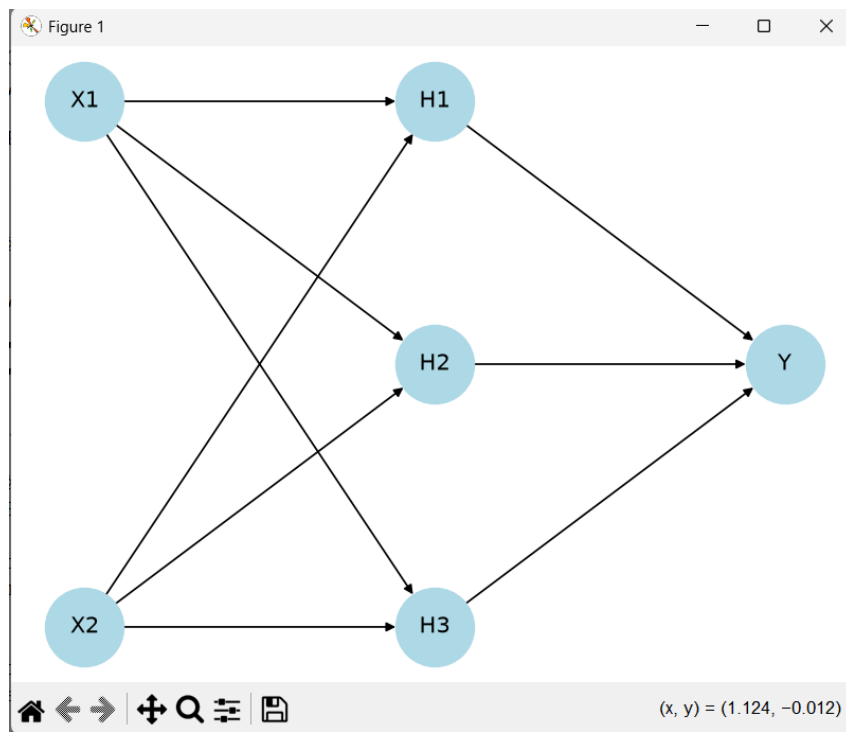
Input and Output

X1 = 1

X2 = 1

Input: [1, 1]

Predicted Output: 1



Result:

The Feed Forward Neural Network was successfully implemented using Python. The neural network was trained using the given input features and target values, and it successfully predicted the output for new user-provided input.